

Salam Baybars

Full-Stack & Systems Developer

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Full-Stack & Systems Developer and 42 Amman graduate with strong experience in React, TypeScript, .NET, PostgreSQL, and real-time application development. Built production-ready full-stack platforms end-to-end using modern web technologies while applying low-level systems programming knowledge gained through intensive C and C++ projects at 42 School. Passionate about backend architecture, networking, multithreading, and scalable software systems, with hands-on experience building a Unix shell, HTTP server, and real-time multiplayer applications from scratch.

PROJECTS

Andary: Full-Stack Multiplayer Educational Platform

Mar 2026

- Built a real-time multiplayer educational platform designed for Tawjihi students across the Arab world using React, TypeScript, Tailwind CSS, .NET, and PostgreSQL.
- Implemented live multiplayer gameplay architecture using WebSockets, allowing players to join public or private rooms and compete in synchronized quiz sessions.
- Developed secure authentication and session management using JWT, refresh tokens, HTTP-only cookies, and Google OAuth 2.0 integration.
- Engineered a scalable backend system for room management, real-time game state synchronization, scoring logic, and multiplayer interactions.
- Designed multilingual support including Arabic RTL, English, and Chinese with responsive UI/UX optimized for desktop and mobile devices.
- Managed deployment and infrastructure configuration using Docker, Nginx, and modern cloud hosting workflows.
- **Demo:** [Andary Live Demo](#)

Webserv: HTTP/1.1 Server from Scratch

Aug 2025

- Built an HTTP/1.1 web server in C++98 using raw POSIX sockets and non-blocking I/O.
- Implemented non-blocking I/O multiplexing using poll() to efficiently handle multiple simultaneous client connections.
- Added CGI execution, static file serving, file uploads, and custom error pages.
- Developed an Nginx-inspired configuration parser supporting routes and server blocks.
- Worked with TCP/IP networking, HTTP internals, and Linux systems programming concepts.

MiniShell: Unix Shell Recreation

Feb 2025

- Recreated a functional Unix shell in C with support for pipes and redirections.
- Implemented command parsing, environment variables, and built-in shell commands.
- Managed process execution, signals, and file descriptors in Linux environments.
- Built lexer and parser logic for shell command interpretation.
- Worked deeply with Unix internals and low-level systems programming concepts.

Inception: Docker Infrastructure Project

Aug 2025

- Built a complete Docker-based infrastructure using Docker Compose and Nginx.
- Configured multiple isolated containers for backend services, databases, and web servers.
- Managed networking, volumes, and environment configuration between containers.
- Implemented secure service communication and reverse proxy configuration with Nginx.
- Worked deeply with Linux, containerization, and deployment workflows.

Cub3D: Raycasting Graphics Engine

Jul 2025

- Developed a 3D graphics engine inspired by Wolfenstein using C and MiniLibX.
- Implemented raycasting algorithms, texture rendering, and player movement systems.
- Built custom map parsing and collision detection logic.
- Managed real-time rendering and keyboard interaction handling.
- Worked with graphics programming and low-level memory management concepts.

World Clock App

Oct 2025

- Built a responsive world clock application using HTML, CSS, and vanilla JavaScript with real-time timezone synchronization.
- Implemented stopwatch, countdown timer, session history tracking, and dark/light theme functionality using modular ES6 architecture.
- Designed a mobile-friendly UI with localStorage persistence and dynamic timezone handling using the JavaScript Intl API.

TECHNICAL SKILLS

- **Programming Languages:**
 - C, C++, TypeScript, JavaScript, Java, SQL
- **Backend & Frameworks:**
 - .NET / ASP.NET, Node.js, REST API Design, JWT Authentication, WebSockets
 - OOP, Data Structures & Algorithms, Systems Programming
- **Databases:**
 - PostgreSQL, MySQL, MariaDB
 - Database Design, Query Optimization
- **Systems & Networking:**
 - TCP/IP, POSIX Sockets, HTTP/1.1 Internals, Non-blocking I/O
 - Multithreading, Mutex Synchronization, Process & Signal Handling
- **Frontend:**
 - React, Vite, Tailwind CSS, HTML5, CSS3
 - Responsive Design, RTL Support
- **DevOps & Tools:**
 - Docker, Docker Compose, Git, Linux, Bash, Nginx, VS Code, WSL

EDUCATION

World Islamic Sciences University

October 2023 – Present

Bachelor's Degree in Computer Science

42 Network (42 Amman)

August 2024 – April 2026

Core Curriculum Diploma in Software Engineering

Peer-to-peer, project-based program focused on systems programming, algorithms, software engineering, and networking through intensive C and C++ projects in Unix/Linux environments.

42 Network (42 Amman)

June 2024 – August 2024

Piscine: Low-Level Programming

Intensive selection program focused on C programming, algorithms, problem solving, and Unix fundamentals.

LANGUAGES

- **Arabic** - Native proficiency
- **English** - Advanced proficiency